UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Important points

Definitions, statements, key points, "why" reasoning and exam traps, topic by topic.

Class 005 · Homogeneous functions and Euler's theorem · compiled 14 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S005·6	Deck 005, slide 6 (rendered .pptx)
T005 00:08:57	AV-summary transcript 005, segment start
CW U1 P20	Unit 1 class-work problem 20
ESA Jan-24 s3 1c	Past paper: Jan 2024, set 3, question 1.c

Contents

1. 1 · Homogeneous functions
2. 2 · Euler's theorem
3. 3 · Second and higher order: the corollary
4. 4 · Extension: when f(u) is homogeneous
5. 5 · Applications of homogeneous functions
6. Past-paper questions

1 Homogeneous functions

Definitions

Homogeneous function (two variables). A function $u = f(x, y)$ is said to be a homogeneous function of degree n if it can be expressed in the form $x^n g\left(\frac{y}{x}\right)$ or $y^n g\left(\frac{x}{y}\right)$, g being an arbitrary function. S005·3 · T005 00:02:45

Homogeneous function (three variables). $u = f(x, y, z)$ is homogeneous of degree n if it can be expressed as $x^n g\left(\frac{y}{x}, \frac{z}{x}\right)$, or $y^n g\left(\frac{x}{y}, \frac{z}{y}\right)$, or $z^n g\left(\frac{x}{z}, \frac{y}{z}\right)$. S005·4 · T005 00:04:38

Scaling test. f is homogeneous of degree n exactly when $f(tx, ty) = t^n f(x, y)$ for all $t > 0$: scaling every variable by t scales the function by t^n. ADDED FOR CLARITY S005·5

Degree. The power n in $x^n g(y/x)$; for a polynomial whose terms all have the same total degree, that common degree. T005 00:01:31

Key points

1. For a polynomial, check that every term has the same total degree; that is the degree ($x^3 + 3x^2y + 3xy^2 + y^3$: degree 3). ESA ★★ T005 00:01:15 · T005 00:01:31
2. To test any function: take out x^n (or y^n) and check that what remains depends only on y/x (and z/x). ESA ★★ S005·3 · T005 00:05:42
3. Either form, $x^n g(y/x)$ or $y^n h(x/y)$, may be used — whichever is convenient — and both give the same degree. ESA ★★ T005 00:04:01
4. Inverse trigonometric or exponential functions of a homogeneous expression are usually not homogeneous; first find an f with $f(u)$ homogeneous (topic 4). ESA ★★★ T005 00:01:47 · T005 00:02:04
5. Homogeneous functions appeared earlier in homogeneous differential equations, solved by $y = vx$. ESA ★ (transcript only) T005 00:00:22

Exam traps

Exam trap. The lecture says the converse holds: "a function of degree 0 should be a constant function". It does not: y/x, $\sin^{-1}(y/x)$ and $\frac{x-y}{x+y}$ all have degree 0 and are not constant.
(added for clarity) T005 00:07:11

Exam trap. "Homogeneous" does not mean "smooth everywhere" (as said at T005 00:11:06); it is a scaling property. $(\sqrt{x} + \sqrt{y})/(x + y)$ is homogeneous but not defined at the origin.
(added for clarity) T005 00:11:06

2 Euler's theorem

Statements in the class material

> *"Statement 1: If $u = f(x, y)$ is a homogeneous function of degree n then $x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} = nu$."* S005·6

> *"Statement 2: If $u = f(x, y, z)$ is a homogeneous function of degree n then $x\dfrac{\partial u}{\partial x} + y\dfrac{\partial u}{\partial y} + z\dfrac{\partial u}{\partial z} = nu$."* S005·6

Key points

1. You do not need to differentiate: check that u is homogeneous, find n, and write $xu_x + yu_y = nu$. ESA ★★★ T005 00:09:17 · T005 00:17:32
2. The lecturer calls Euler's theorem "the most important one" of the session. ESA ★★★ T005 00:08:57
3. Check: $u = x^2 + y^2$ gives $x(2x) + y(2y) = 2u$, degree 2. ESA ★★ T005 00:10:31
4. Meaning: the change along x scaled by x, plus the change along y scaled by y, equals the function scaled by its degree. ESA ★ T005 00:09:46 · T005 00:11:24

Why?

Why is $xu_x + yu_y = nu$? (A short proof.)
Differentiate $u(tx, ty) = t^n u(x, y)$ with respect to t using the chain rule: $x\,u_x(tx, ty) + y\,u_y(tx, ty) = nt^{n-1}u(x, y)$. Put $t = 1$.

ADDED FOR CLARITY S005·6

3 Second and higher order: the corollary

Statements in the class material

> *"Corollary 1: If $u = f(x, y)$ is a homogeneous function of degree n then $x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} = n(n-1)u$."* S005·6

Key points

1. The mixed partials u_{xy} and u_{yx} are equal here (continuous derivatives), which is why the middle term is $2xy\,u_{xy}$. ESA ★★ T005 00:12:04
2. For $u = u_1 + u_2$ with different degrees n_1, n_2, apply the results to each part: $x^2u_{xx} + \cdots = n_1(n_1 - 1)u_1 + n_2(n_2 - 1)u_2$. ESA ★★★ ADDED FOR CLARITY ESA Jan-24 s3 1c · CW U1 P22
3. Extend to three or more variables as an exercise; the lecture leaves it open and points to a research paper on higher-order extensions. ESA ★ S005·6 · T005 00:14:59

Why?

Why is the second-order right side $n(n-1)u$?
u_x and u_y are homogeneous of degree $n - 1$, so Euler's theorem applied to them gives $xu_{xx} + yu_{xy} = (n-1)u_x$ and $xu_{yx} + yu_{yy} = (n-1)u_y$. Multiply by x and y, add, and use $xu_x + yu_y = nu$.

ADDED FOR CLARITY S005·6

4 Extension: when f(u) is homogeneous

Statements in the class material

> *"If $f(u)$ be a homogeneous function of degree n in x and y, then* $x\dfrac{\partial u}{\partial x}+y\dfrac{\partial u}{\partial y}=n\dfrac{f(u)}{f'(u)}$ *and* $x^2\dfrac{\partial^2 u}{\partial x^2}+2xy\dfrac{\partial^2 u}{\partial x\partial y}+y^2\dfrac{\partial^2 u}{\partial y^2}=g(u)[g'(u)-1]$ *where* $g(u)=n\dfrac{f(u)}{f'(u)}$*."* S005·7

Key points

1. Spot the extension case: $u=\sin^{-1}(\dots)$, $\tan^{-1}(\dots)$, $\log(\dots)$ or $e^{(\cdots)}$ of a homogeneous expression; then f is $\sin$, $\tan$, $\exp$ or $\log$ respectively. ESA ★★★ (added for clarity) S005·7 · ESA May-24 1a
2. The degree n belongs to $f(u)$, not to u. ESA ★★★ S005·7

Why?

Where does $n\,f(u)/f'(u)$ come from?
Apply Euler to $F=f(u)$: $xF_x+yF_y=nF$. By the chain rule $F_x=f'(u)u_x$, $F_y=f'(u)u_y$, so $f'(u)(xu_x+yu_y)=nf(u)$.
(added for clarity) S005·7

Exam traps

Exam trap. For $u=\tan^{-1}\frac{x^4+y^4}{xy}$ the degree is that of $\tan u=\frac{x^4+y^4}{xy}$, i.e. $4-2=2$, not of u.
(added for clarity) ESA May-24 1a

5 Applications of homogeneous functions

Definitions

Extensive / intensive variable. Extensive variables (mass, volume, energy) double when the system is doubled: homogeneous of degree 1. Intensive variables (pressure, temperature, density) do not change: degree 0. (added for clarity) S005·5 · T005 00:16:26

Statements in the class material

> *"A constant function is homogeneous of degree 0. Linear functions are homogeneous of degree 1. The degree of the homogeneous function can be positive, negative or zero."* S005·5

Key points

1. A constant function is homogeneous of degree 0; linear functions are homogeneous of degree 1. ESA ★ S005·5 · T005 00:06:52
2. Euler's theorem also describes the scaling of the function: $n\,u$. ESA ★ S005·5 · T005 00:15:57
3. Applications: production functions, marginal productivities and elasticity of demand in economics; extensive (degree 1) and intensive (degree 0) variables in thermodynamics. ESA ★ S005·5 · T005 00:16:09
4. Any function of ratios only, such as $f(x/y,\, y/z,\, z/x)$, has degree 0, so $xu_x + yu_y + zu_z = 0$ (ESA Jan 2024 set 1). ESA ★★★ (added for clarity) ESA Jan-24 s1 1d

Exam traps

Exam trap. The lecture swaps the examples: mass is extensive (degree 1); pressure and temperature are intensive (degree 0). (added for clarity) T005 00:16:46

★ Past-paper questions

Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024. Euler's theorem appears in six of them: Jul 2023 Q1.a, May 2024 Q1.a, June 2024 Q1.b (the same question as May 2024), Jan 2024 set 3 Q1.c, set 4 Q1.d, and set 1 Q1.d (a degree-0 function, also solvable by the chain rule). Five are 6-mark questions, so this is one of the most examined topics in Unit 1. The definition, applications and third-order result were not asked.

3 · Second and higher order: the corollary

ESA Jul 2023 (6 marks) UE22MA141A Q1.a · solved in W3.1

If $u(x,y) = x^3y^2 \sin^{-1}\left(\frac{y}{x}\right)$, prove that (i) $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 5u$, (ii) $x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} = 20u$.

ESA Jan 2024 (6 marks) UE23MA141A set 3, Q1.c · solved in W3.5

If $u = x^3 \sin^{-1}\left(\frac{y}{x}\right) + x^4 \tan^{-1}\left(\frac{y}{x}\right)$, find $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} + xu_x + yu_y$ at $x = 1$, $y = 1$ by using Euler's theorem and its extension.

ESA Jan 2024 (6 marks) UE23MA141A set 4, Q1.d · solved in W3.4

If $u = x^3 \sin^{-1}\left(\frac{y}{x}\right) + y^{-3} \tan^{-1}\left(\frac{x}{y}\right)$, evaluate $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} + xu_x + yu_y$ by using Euler's theorem and its extension.

4 · Extension: when f(u) is homogeneous

ESA May 2024 (6 marks) UE22MA141A Q1.a · solved in W4.1

If $u = \tan^{-1}\left(\frac{x^4+y^4}{xy}\right)$, prove that (i) $xu_x + yu_y = \sin 2u$, (ii) $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = \sin 2u(1 - 4\sin^2 u)$.

ESA June 2024 (6 marks) UE23MA141A Q1.b · solved in W4.1

Same question as May 2024 Q1.a: $u = \tan^{-1}\left(\frac{x^4+y^4}{xy}\right)$, prove (i) and (ii).

5 · Applications of homogeneous functions

ESA Jan 2024 (5 marks) UE23MA141A set 1, Q1.d · solved in W5.1

If $u = f\left(\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\right)$, find $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z}$ (degree 0; also solvable by the chain rule).